

Project Integration

Pothole Detection and Reporting Platform Team

May 22, 2026

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Final PDF Submission

This document is organized in the same order as the Project Integration example and contains all required artifacts from the repository.

1. README.md File

Source: ../README.md

Pothole

2. Working Agreement

Source: ../docs/WORKING_AGREEMENT.md

Working Agreement

Meeting Information

- Weekly meeting day/time: Tuesdays at 7:00 PM (America/New_York)
- Meeting platform: Discord voice channel
- Backup platform: Microsoft Teams

Communication Expectations

- Preferred channels: Discord for daily communication, GitHub comments for issue/PR context
- Expected response time: within 12 hours on weekdays, within 24 hours on weekends
- Urgent blockers: tag '@here' in Discord and open a GitHub issue immediately

Accountability Rules

- Missing meetings without notice: responsible for publishing next meeting notes and action summary
- Repeated lateness (3+ times): temporary reduction in planning vote priority for that sprint
- Lack of participation: task reassignment plus documented recovery plan
- Failure to respond to teammates: escalation to team lead, then professor if unresolved

GitHub Workflow Agreement

- No direct pushes to 'main'; all changes must go through Pull Requests
- Minimum 1 reviewer approval required before merge
- PR must include linked issue and checklist completion
- Branch naming convention:
 - 'hardware/<short-description>' for CAD or electronics integration work
 - 'firmware/<short-description>' for embedded code updates
 - 'software/<short-description>' for dashboard/backend/frontend updates
 - 'docs/<short-description>' for documentation updates

Testing Requirements

- Before merge, contributor must verify acceptance criteria from linked issue
- Hardware changes:
 - CAD fit check completed
 - IoT module placement and mounting constraints verified
- Firmware changes:
 - Build/flash verification completed
 - Sensor read + telemetry transmission validated on test device
- Software changes:
 - Relevant unit/integration tests pass
 - Manual smoke test of impacted dashboard flow completed
- Reviewer is responsible for confirming testing evidence is included in PR

3. Current Changelog

Source: ../docs/CHANGELOG.md

Changelog

All notable changes to this project will be documented in this file.

2026-05-21

Added

- Created initial project documentation structure in local repository.
- Added 'README.md' with team table and non-technical project overview.
- Added working agreement at 'docs/WORKING_AGREEMENT.md'.
- Added project backlog templates and issue drafts for GitHub import.
- Added planning artifacts for milestones and Kanban board setup.

Links

- Pull Request: [PR #1](https://github.com/your-org/your-repo/pull/1)

4. Project Backlog Using GitHub Issues

The following backlog content was used to create project issues.

Source: ../issues/BACKLOG.md

Backlog Draft (User Story Titles)

Fully Written Issues (ready now)

1. As a Hardware Engineer, I want a CAD enclosure that fits the sensor and IoT modules so that the deployment is successful.
2. As a Firmware Developer, I want embedded firmware to collect accelerometer and GPS data so that road conditions can be monitored.
3. As a Data Reviewer, I want a dashboard that visualizes pothole events on a map and timeline so that trends can be identified.
4. As a Project Integrator, I want a validated hardware-firmware-software data pipeline so that sensor data is reliable.

Remaining Story Titles (expand later)

1. As a Firmware Developer, I want configurable sampling rates so that battery usage can be optimized.
2. As a Hardware Engineer, I want vibration-resistant mounting features in the CAD design so that modules are protected.
3. As an Analyst, I want event filtering by severity and date range so that trends are easier to analyze.
4. As a Team Member, I want deployment logs linked to each test run so that field debugging is traceable.

5. Fully Written Issue Content

The following file contains the fully written issue details for each student.

Source: ../issues/FULLY_WRITTEN_ISSUES.md

Example Fully Written Issues

Issue 1

Title: As a Hardware Engineer, I want a CAD enclosure that fits the sensor and IoT modules so that the

Description

Design and validate a CAD enclosure that houses the microcontroller, sensor package, and communication

Acceptance Criteria

- [] CAD model includes mounting points for all selected modules.
- [] Enclosure dimensions satisfy installation space constraints.
- [] Design includes cable routing and access for maintenance.
- [] Exported files are ready for prototype fabrication.

Testing Criteria

- [] Dry-fit test confirms module alignment and clearance.
- [] Vibration simulation or manual shake test shows no loose components.
- [] Assembly can be completed and reopened without component damage.

Notes

Assign to milestone: Hardware Integration.

Issue 2

Title: As a Firmware Developer, I want embedded firmware to collect accelerometer and GPS data so that

Description

Implement firmware tasks for sampling motion/location data, deriving pothole events from thresholds, and

Acceptance Criteria

- [] Firmware reads accelerometer and GPS data at configured intervals.
- [] Pothole event logic is triggered from defined thresholds.
- [] Event payload includes timestamp, location, and severity score.
- [] Device retries transmission on temporary network failures.

Testing Criteria

- [] Firmware builds and flashes successfully to target board.
- [] Bench test with simulated vibration triggers event creation.
- [] Confirm event payload is received by software ingestion endpoint.

Notes

Assign to milestone: Firmware Data Pipeline.

Issue 3

Title: As a Data Reviewer, I want a dashboard that visualizes pothole events on a map and timeline so th

Description

Create a software dashboard that displays ingested events with map markers, severity coloring, and time

Acceptance Criteria

- [] Dashboard lists events with location, timestamp, and severity.
- [] Map view renders event pins with severity-based colors.
- [] Timeline/date filter updates visualizations correctly.
- [] User can open an event detail panel from map or list.

Testing Criteria

- [] UI renders correctly with sample and live event data.
- [] Filters return consistent counts across list and map views.
- [] Basic responsiveness validated on desktop and mobile widths.

Notes

Assign to milestone: Software Visualization.

Issue 4

Title: As a Project Integrator, I want a validated hardware-firmware-software data pipeline so that sen

Description

Run end-to-end integration tests from sensor trigger to dashboard display and document failure points w

Acceptance Criteria

- [] A generated field event appears in dashboard within target latency.
- [] Failed transmissions are retried and eventually resolved.
- [] Integration test steps are documented for repeatability.
- [] Known limitations are captured in project notes.

Testing Criteria

- [] At least 5 successful end-to-end event runs are recorded.
- [] One simulated network interruption test is completed.
- [] Evidence (logs/screenshots) attached to issue or PR.

Notes

Assign to milestone: End-to-End Validation.

6. Issue Screenshots

One screenshot of a fully written issue per student. Each screenshot shows issue title, issue body, and assigned milestone (if applicable).

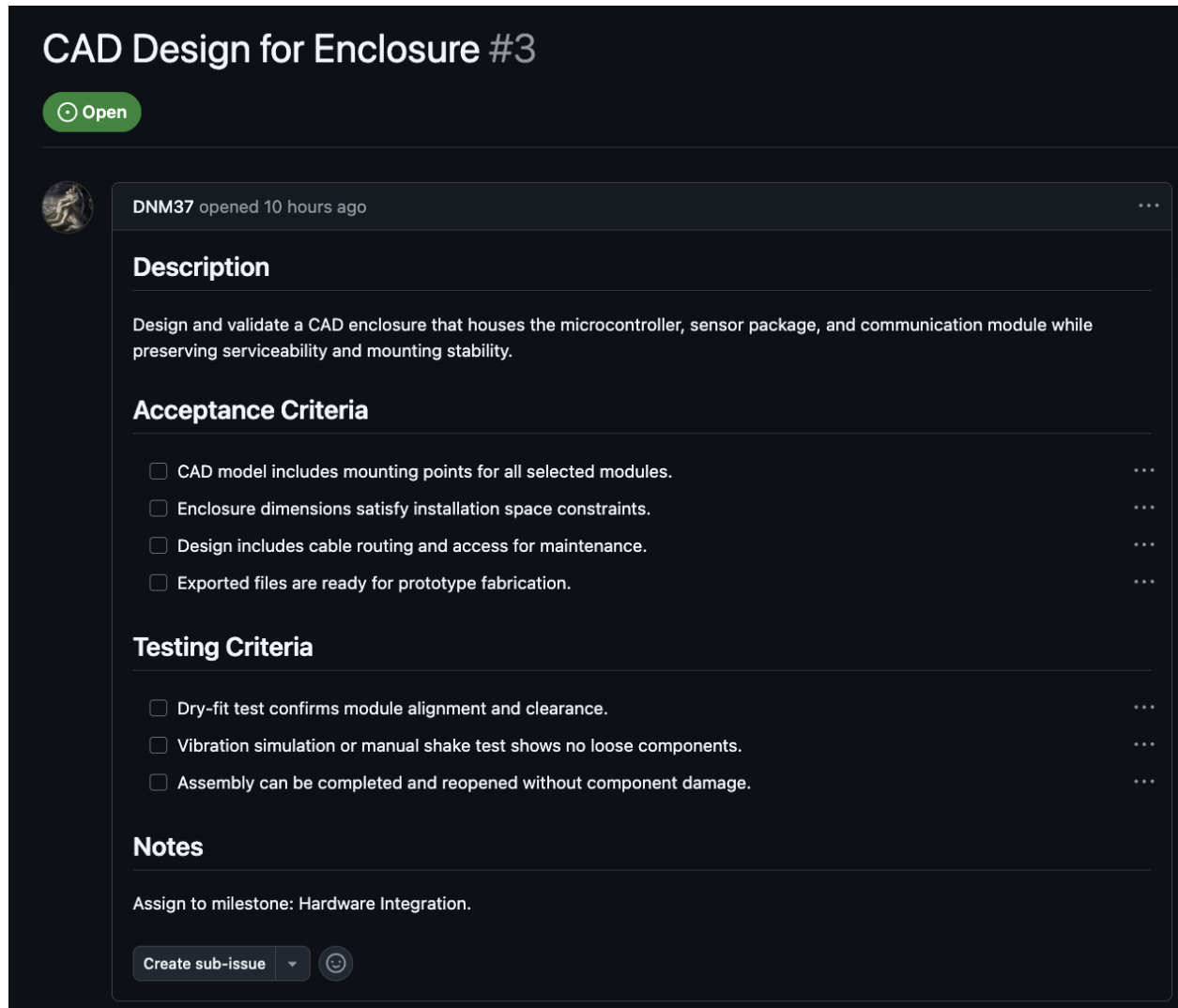


Figure 1: Student 1 Fully Written Issue

Implement sensor sampling for motion and location data #1

Open



sthapanavichet opened 11 hours ago

Last edited by sthapanavichet ▾ ⋮

Description

Implement firmware tasks to read accelerometer and GPS data at configured intervals for pothole detection.

Acceptance Criteria

- ☐ Firmware reads accelerometer data at configured intervals. ⋮
- ☐ Firmware reads GPS data at configured intervals. ⋮
- ☐ Sensor sampling intervals are configurable. ⋮
- ☐ Sampled data is available to downstream pothole detection logic. ⋮

Testing Criteria

- ☐ Firmware builds successfully. ⋮
- ☐ Firmware flashes successfully to the target board. ⋮
- ☐ Confirm accelerometer readings are captured during bench testing. ⋮
- ☐ Confirm GPS/location readings are captured or mocked successfully. ⋮

Milestone

Firmware Data Pipeline

Create sub-issue ▾



Figure 2: Student 2 Fully Written Issue

Dashboard #5

 Open



Atem2222 opened 54 minutes ago

Last edited by Atem2222 ▾ ⋮

Description

Create a software dashboard that displays ingested events with map markers, severity coloring, and timeline filtering.

Acceptance Criteria

☐ Dashboard lists events with location, timestamp, and severity.

⋮

☐ Map view renders event pins with severity-based colors.

⋮

☐ Timeline/date filter updates visualizations correctly.

⋮

☐ User can open an event detail panel from map or list.

⋮

Testing Criteria

☐ UI renders correctly with sample and live event data.

⋮

☐ Filters return consistent counts across list and map views.

⋮

☐ Basic responsiveness validated on desktop and mobile widths.

⋮

Notes

Assign to milestone: Software Visualization.

Create sub-issue ▾



Figure 3: Student 3 Fully Written Issue

Validated hardware-firmware-software data pipeline integration #6

Open



AmirAbdalla opened 50 minutes ago

...

User Story

As a Project Integrator, I want a validated hardware-firmware-software data pipeline so that sensor events appear reliably in the dashboard.

Description

Run end-to-end integration tests from sensor trigger to dashboard display and document failure points with corrective actions.

Acceptance Criteria

- ☐ A generated field event appears in dashboard within target latency. ...
- ☐ Failed transmissions are retried and eventually resolved. ...
- ☐ Integration test steps are documented for repeatability. ...
- ☐ Known limitations are captured in project notes. ...

Testing Criteria

- ☐ At least 5 successful end-to-end event runs are recorded. ...
- ☐ One simulated network interruption test is completed. ...
- ☐ Evidence (logs/screenshots) attached to issue or PR. ...

Create sub-issue



Figure 4: Student 4 Fully Written Issue

7. Milestones

Source: ../planning/MILESTONES.md

Milestones Plan

Create these milestones in GitHub and assign issues accordingly.

1. Hardware Integration

- Scope: CAD enclosure design, IoT module fit, mounting validation
- Example linked issues:
 - As a Hardware Engineer, I want a CAD enclosure that fits the sensor and IoT modules so that the dep

2. Firmware Data Pipeline

- Scope: embedded sensing, event logic, transmission reliability
- Example linked issues:
 - As a Firmware Developer, I want embedded firmware to collect accelerometer and GPS data so that roa

3. Software Visualization

- Scope: data ingestion display, map view, timeline filtering
- Example linked issues:
 - As a Data Reviewer, I want a dashboard that visualizes pothole events on a map and timeline so that

4. End-to-End Validation

- Scope: integrated system validation and reliability checks
- Example linked issues:
 - As a Project Integrator, I want a validated hardware-firmware-software data pipeline so that sensor

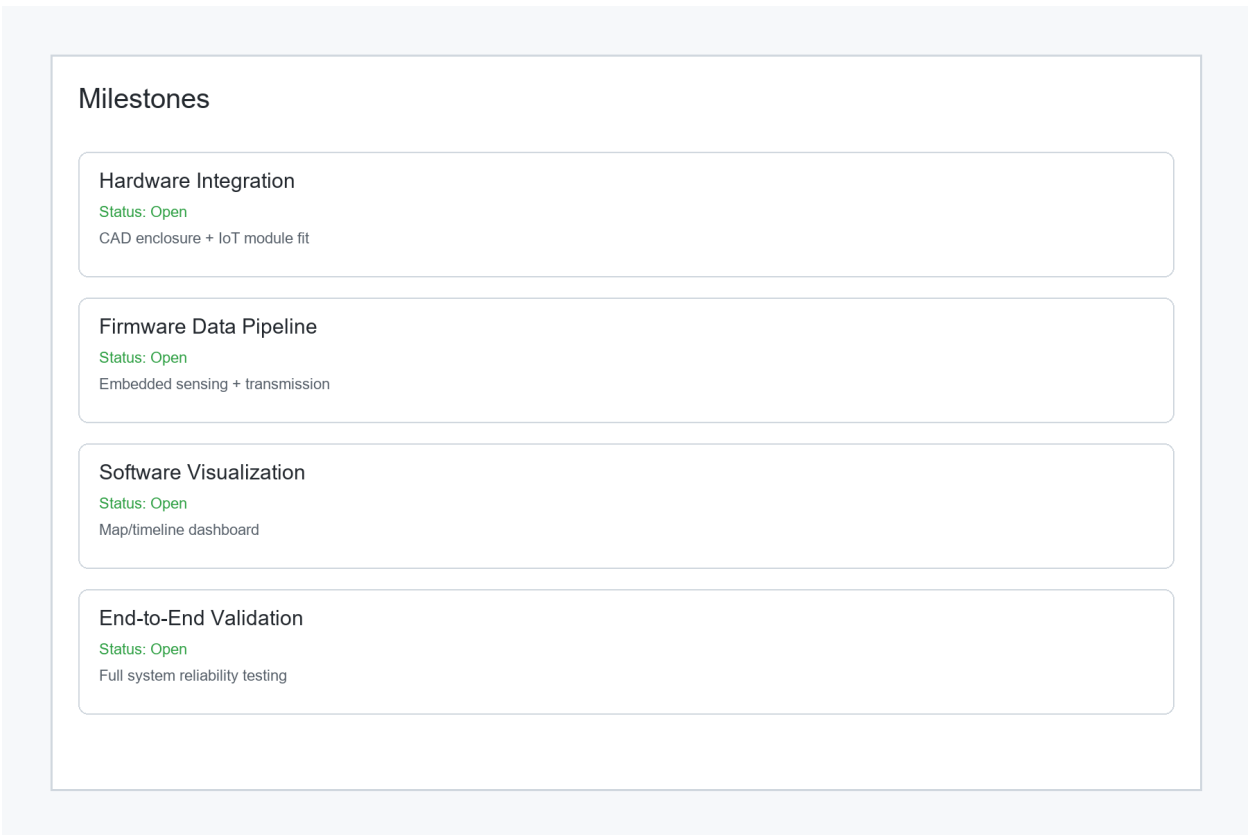


Figure 5: GitHub Milestones View

8. Kanban Board

Source: ../planning/KANBAN.md

Kanban Board Setup

Suggested columns:

- To Do
- In Progress
- Reviewing
- Done

Yes, the board should be populated with actual issues. The assignment says issues should be placed in a

Initial population suggestion:

- 'To Do'
 - CAD enclosure fit and mounting design
 - Firmware sensor sampling task
 - Dashboard timeline filter implementation
- 'In Progress'
 - IoT module integration into CAD shell
- 'Reviewing'
 - Event payload schema definition PR
- 'Done'
 - Repository setup and project integration docs

Usage rules:

- New issues enter 'To Do'.
- When a team member starts work, move to 'In Progress'.
- When a PR is open, move to 'Reviewing'.
- After merge and verification, move to 'Done'.

Board hygiene:

- Every card must have an assignee.
- Every card must be linked to an issue.
- Every card in 'Reviewing' must reference an open PR.

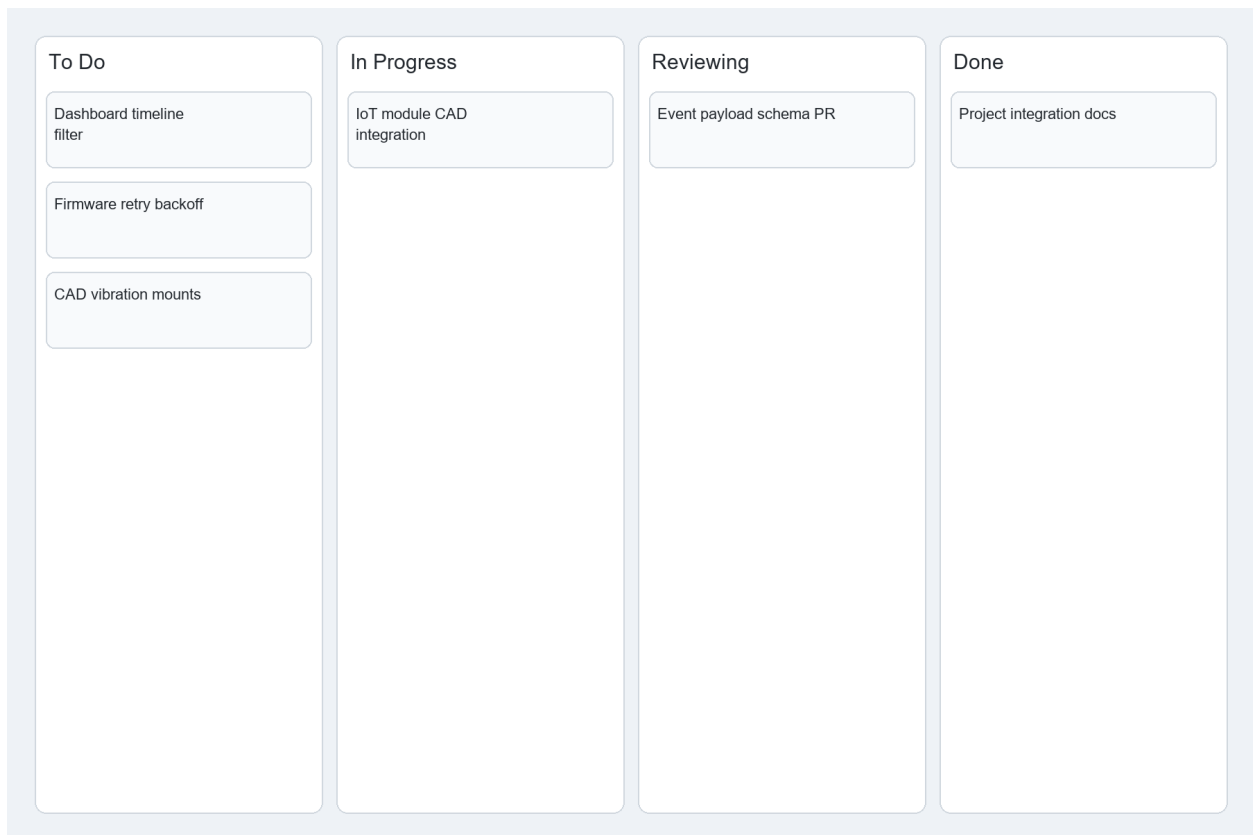


Figure 6: GitHub Kanban Board View

Appendix: Submission Checklist

Source: ../SUBMISSION_CHECKLIST.md

Project Integration Submission Checklist

1. README includes team table and 2-3 paragraph non-technical description.
2. 'docs/WORKING_AGREEMENT.md' includes all required sections.
3. 'docs/CHANGELOG.md' includes date, short change description, and PR link.
4. GitHub issues created from 'issues/BACKLOG.md'.
5. Fully written issue per student includes template sections.
6. Milestones created and issues assigned.
7. Kanban board created with required columns.
8. Final PDF assembled with required screenshots.